# COMPACTIFICATION DESCRIPTION OF F_! ALONG AN OPEN IMMERSION INTO A PROPER SCHEME

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

*Editorial note.* Editorial note: two light steps are asserted rather than displayed: the reduction to support containment is justified by the explicit nature of the covariance remark map, and the final support statement is called immediate from the vanishing of the stalks of j_!F at geometric points of the complement; both are definitional given the displayed data.

*Allowed standard prerequisites (editorial).*

- Agreement of lower shriek with pushforward for proper morphisms (Lemma lemma-proper-f-shriek, canonical tag 0F51). cited at statement lines 540-541. Gives the final identification f_bar_! = f_bar_* for the proper compactification.
- Restriction of j_! sheaves (Etale Cohomology, Lemma etale-cohomology-lemma-jshriek-open, canonical tag 0F70). cited at proof lines 547-548. (j_!F)|_X = F, identifying the displayed arrow as the covariance map.
- Covariance of f_! for separated morphisms (Remark remark-covariance-f-shriek-separated, canonical tag 0F53). cited at proof lines 549-552. Supplies the injective map f_!(G|_X) -> f_bar_!G whose explicit form drives the reduction.

*Source reference.* Original lemma 0F51: sources/stacks-chow/tags.txt, source lines 18921–18921. Cited at statement lines 540-541: gives the final identification \overline{f}_! = \overline{f}_* for the proper compactification.

*Source reference.* Original lemma 0F70: sources/stacks-chow/tags.txt, source lines 18990–18990. Cited at proof lines 547-548: (j_!F)|_X = F, identifying the displayed arrow as the covariance map.

*Source reference.* Original remark 0F53: sources/stacks-chow/tags.txt, source lines 18923–18923. Cited at proof lines 549-552: supplies the injective map f_!(G|_X) -> \overline{f}_!G whose explicit form drives the reduction.

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**Lemma 0.2.** *Let $Y$ be a scheme. Let $j : X \to \overline{X}$ be an open immersion of schemes over $Y$ with $\overline{X}$ proper over $Y$. Denote $f : X \to Y$ and $\overline{f} : \overline{X} \to Y$ the structure morphisms. For $\mathcal{F} \in Ab(X_{étale})$ there is a canonical isomorphism (see proof)*

$$f_!\mathcal{F} \longrightarrow \overline{f}_!j_!\mathcal{F}$$

*As we have $\overline{f}_! = \overline{f}_*$ by Lemma 0F51 we obtain $\overline{f}_* \circ j_! = f_!$ as functors $Ab(X_{étale}) \to Ab(Y_{étale})$.*

*Proof.* We have $(j_!\mathcal{F})|_X = \mathcal{F}$, see Étale Cohomology, Lemma 0F70. Thus the displayed arrow is the injective map $f_!(\mathcal{G}|_X) \to \overline{f}_!\mathcal{G}$ of Remark 0F53 for $\mathcal{G} = j_!\mathcal{F}$. The explicit nature of this map implies that it now suffices to show: if $V \in Y_{étale}$ and $s \in \overline{f}_!\mathcal{G}(V) = \overline{f}_*\mathcal{G}(V) = \mathcal{G}(\overline{X}_V)$ is a section, then the support of $s$ is contained in the open $X_V \subset \overline{X}_V$. This is immediate from the fact that the stalks of $\mathcal{G}$ are zero at geometric points of $\overline{X} \setminus X$. □